Guy Ohayon

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I am a Computer Science PhD candidate at the Technion, working with Prof. Michael Elad and Prof. Tomer Michaeli. My research is on the theory and practice of generative image restoration and compression techniques.

Education

Computer Science | *Doctor of Philosophy (direct track)* — March 2021 - Present
Technion

Computer Engineering | *Bachelor of Science* — March 2017 - March 2021
Technion — GPA: 95.4 (Summa Cum Laude)

- Member of the EMET Excellence Program (`https://ece.technion.ac.il/emet-excellence-program/`)

Work Experience

Verily | *Research Scientist Intern* — May 2024 - September 2024
- Applied research in medical imaging

Microsoft | *Research Scientist Intern* — June 2022 - September 2023
- Applied research in computer vision and language models

Amazon Web Services | *Computer Vision Research Intern* — August 2021 - November 2021
- Applied research in text image recognition and self-supervised learning

Google | *Software Engineering Intern* — May 2020 - April 2021
- Designing a scalable TCP packet loss estimation algorithm based on packet sampling

Intel | *Student Machine Learning Engineer* — August 2018 - June 2019
- Using reinforcement learning techniques to compile neural networks on dedicated hardware

Maisha Consulting | *Remote-sensing Systems Integrator* — September 2016 - March 2017
- Integrating cost-effective remote sensing systems to help identify environmental criminals in Africa

Teaching Experience

Technion | *Teaching Assistant, Numerical Algorithms - 2340125* — October 2024 - March 2025

Technion | *Teaching Assistant, Machine Learning - 046195* — October 2021 - March 2022

Technion | *Teaching Assistant, Introduction to Artificial Intelligence - 236501* — March 2021 - July 2021

Honors and Awards

- Technion Artificial Intelligence Hub (Tech AI) Student Research Prize for Cross-PI Collaboration in Data Science in Funding of PBC (VATAT) (2023 - 2024)
- Scholarship for Excellent Graduate Students, Faculty of Computer Science, Technion (2023-2025)
- Commendation (4th place) at the Research Day for Graduate Students, Faculty of Computer Science, Technion (2024)
- Winner (1st place) at the Research Day for Graduate Students, Faculty of Computer Science, Technion (2021)
- Best Student Paper Award, Advances in Image Manipulation (AIM) Workshop, ICCV (2021)
- Amdocs Best Project Contest Award for Undergraduate Students, Faculty of Computer Science, Technion (2021)
- EMET Excellence Program, Faculty of Electrical and Computer Engineering, Technion (2017-2021)
- President's List Award for Excellence ($\times 6$), Technion (2017-2021)
- Dean's List Award for Excellence ($\times 2$), Faculty of Electrical and Computer Engineering, Technion (2017-2021)
- Head of Military Intelligence Directorate Prize for Creative Thinking (2016)

List of Publications

[1] **Guy Ohayon**, Hila Manor, Tomer Michaeli, and Michael Elad. Compressed image generation with denoising diffusion codebook models. *Under review*. URL https://arxiv.org/abs/2502.01189.

[2] **Guy Ohayon**, Tomer Michaeli, and Michael Elad. Posterior-mean rectified flow: Towards minimum MSE photo-realistic image restoration. In *The 13th International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=hPOt3yUXii.

[3] **Guy Ohayon**, Michael Elad, and Tomer Michaeli. Perceptual fairness in image restoration. In *The 38th Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024. URL https://openreview.net/forum?id=M2QREVHK1V.

[4] **Guy Ohayon**, Tomer Michaeli, and Michael Elad. The perception-robustness tradeoff in deterministic image restoration. In *The 41st International Conference on Machine Learning (ICML)*, 2024. URL https://proceedings.mlr.press/v235/ohayon24a.html.

[5] **Guy Ohayon**, Theo Adrai, Michael Elad, and Tomer Michaeli. Reasons for the superiority of stochastic estimators over deterministic ones: Robustness, consistency and perceptual Quality. In *The 40th International Conference on Machine Learning (ICML)*, 2023. URL https://proceedings.mlr.press/v202/ohayon23a.html.

[6] **Guy Ohayon**, Theo Adrai, Gregory Vaksman, Michael Elad, and Peyman Milanfar. High perceptual quality image denoising with a posterior sampling cgan. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, 2021. URL https://openaccess.thecvf.com/content/ICCV2021W/AIM/html/Ohayon_High_Perceptual_Quality_Image_Denoising_With_a_Posterior_Sampling_CGAN_ICCVW_2021_paper.html.

[7] Theo Adrai, **Guy Ohayon**, Michael Elad, and Tomer Michaeli. Deep optimal transport: A practical algorithm for photo-realistic image restoration. In *The 37th Conference on Neural Information Processing Systems (NeurIPS)*, 2023. URL https://openreview.net/forum?id=bJJY9TFfe0.

[8] Sean Man, **Guy Ohayon**, Theo Adrai, and Michael Elad. High-Perceptual Quality JPEG Decoding via Posterior Sampling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2023. URL https://openaccess.thecvf.com/content/CVPR2023W/NTIRE/html/Man_High-Perceptual_Quality_JPEG_Decoding_via_Posterior_Sampling_CVPRW_2023_paper.html.

[9] Sean Man, **Guy Ohayon**, Ron Raphaeli, and Michael Elad. Proxies for distortion and consistency with applications for real-world image restoration. *Under review*. URL https://arxiv.org/abs/2501.12102.